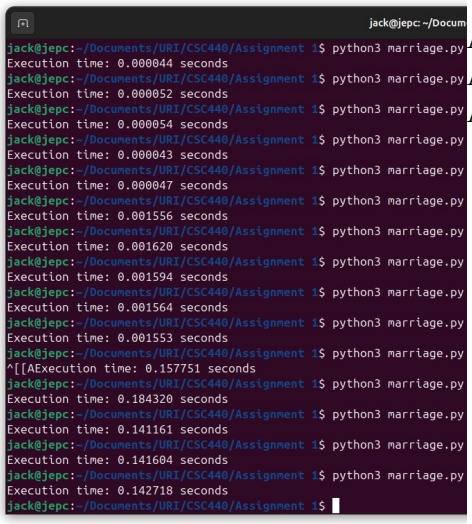
Part 1: Perfect Marriage

Jonathan Pincince & Shina Adewumi



Avg for 10: 0.000048s → 4.8 * 10^{-6} s per input
Avg for 100: 0.0015774s → 1.5774 * 10^{-5} s per input
Avg for 1000: 0.767554s → 7.67554 * 10^{-4} s per input

Since the complexity of the algorithm is O(n^2), it makes sense that the runtime per input gets slower/worse by an order of magnitude for each power of ten that the input increases. Since n^2 is quadratic, one would expect that a graph of inputs vs. runtime would look like a quadratic (exponential) curve, and this is borne out by the graph below.

